



DESCRIPTION				Labor		Material Cost	Equipment Cost	Total Cost
				MH	Rate			
1. Base Scope - CO2 Chillers Infrastructure								
Discipline	Sheet Number	Reference #	Title Block					
General	GD.01	2	GENERAL COVER SHEET					\$ -
Mechanical	MD.01	2	MECHANICAL COVER SHEET					\$ -
			Provide mobilization, temporary facilities, coordination, and overall project execution requirements; includes general scope but no direct install scope.					\$ -
Mechanical	MD.00	2	MECHANICAL 3RD FL CHILLED WATER PIPING PLAN					\$ -
			Demo existing CHW U-joint and modify existing piping; install new insulated CHWS/CHWR mains, branches, valves, circuit setters, and future stub-outs with supports and seismic bracing.					\$ -
Mechanical	M2.01	2	MECHANICAL 3RD FL CO2 PIPING PLAN					\$ -
			Provide CO2 transfer hose routing infrastructure including unistrut supports, pipe sleeve elbows, and ceiling stub-outs; coordinate with SLAC for hose installation and final connections.					\$ -
Mechanical	M2.02	2	MECHANICAL BASEMENT CO2 PIPING PLAN					\$ -
			Provide CO2 routing support infrastructure at basement including coordination of shaft transitions and layout for SLAC-provided transfer hoses; no direct piping installation by contractor.					\$ -
Mechanical	M2.03	2	MECHANICAL SUB-BASEMENT CO2 PIPING PLAN					\$ -
			Install CO2 sleeve penetrations and pipe sleeve elbows at sub-basement and provide routing infrastructure for new and future transfer hoses by SLAC.					\$ -
Electrical	ED.01	2	ELECTRICAL COVER SHEET					\$ -
			Provide electrical scope including demolition coordination, new power infrastructure requirements, and compliance with SLAC electrical standards.					\$ -
Electrical	ED.02	2	ELECTRICAL EQUIPMENT SCHEDULES					\$ -
			Furnish electrical equipment and feeders per schedules, including contractor-furnished gear coordinated with SLAC-installed components where noted.					\$ -
Electrical	ES.01D	2	ELECTRICAL LCLS B950 R330 DEMO POWER PLAN					\$ -
			Demo existing feeders, panel connections, and associated electrical components to accommodate new CO2 chiller power distribution.					\$ -
Electrical	ES.01	2	ELECTRICAL LCLS B950 R330 POWER PLAN					\$ -
			Install new electrical distribution, feeders, and equipment connections to support CO2 chiller systems and associated loads.					\$ -
Electrical	ES.01D	2	ELECTRICAL SINGLE LINE DIAGRAM DEMO					\$ -
			Remove existing electrical distribution components and connections per demolition single line diagram.					\$ -
Electrical	ES.02	2	ELECTRICAL SINGLE LINE DIAGRAM					\$ -
			Modify and extend existing electrical distribution system per single line diagram requirements.					\$ -
Electrical	ES.03	2	ELECTRICAL SINGLE LINE DIAGRAM NEW					\$ -
			Install new electrical distribution equipment, feeders, and protection devices for CO2 chiller infrastructure.					\$ -
Electrical	ES.04	2	ELECTRICAL PANEL SCHEDULES					\$ -
			Modify panelboards, breakers, and circuiting to accommodate new chiller loads and infrastructure.					\$ -
Electrical	ES.05	2	ELECTRICAL PANEL SCHEDULES					\$ -
			Provide additional panel schedule updates and circuit assignments for new and future CO2 chiller loads.					\$ -
Electrical	EE.01	2	ELECTRICAL DETAILS					\$ -
			Install conduits, supports, terminations, and equipment connections per electrical details.					\$ -
Structural	SD.01	2	STRUCTURAL COVER SHEET AND DETAILS					\$ -
			Provide structural supports, anchorage, and attachment details for piping supports and equipment coordination with existing structure.					\$ -
2. Base Scope - PCW Skid Demo and Relocation								
Mechanical	MD3.0	4	LCLS II - HE-EI BSMT (NEH) - OVERALL PLAN - DEMO					\$ -
			Disconnect, remove, and salvage existing PCW skid, control panel, supports, and associated piping, demo CHWS/CHWR and PCWS/PCWR branches and cap for future tie-ins, coordinate draining, flushing, and controls continuity with SLAC.					\$ -
Electrical	ED3.2	5	LCLS II - HE-EI ELECTRICAL BSMT-PWR DEMO-OVERALL					\$ -
			Demo existing feeders, VFD connections, and associated raceways serving PCW skid equipment.					\$ -
3. Base Scope - PCW System Upgrade								
Mechanical	MD.0	4	LCLS II - HE-EI SYM, LEGENDS, AND GEN. NOTES					\$ -
			Provide general mechanical requirements including coordination, as-built, code compliance, patching, penetration sealing, and demolition handling requirements.					\$ -
Mechanical	ME.0	4	LCLS II - HE-EI MECHANICAL SCHEMATIC - PCW					\$ -
			Demo existing PCW system connections and install new system routing including skid connections, branch distribution, valves, risers, and rebalance system flows per schematic.					\$ -
Mechanical	M7.0	4	LCLS II - HE-EI MECHANICAL MECHANICAL SCHEDULES					\$ -
			Provide piping, equipment capacities, and support spacing per schedules; use for installation requirements but not standalone bid item.					\$ -
3.1 PCW Skid Installation								
Electrical	E1.3	5	LCLS II - HE-EI ELECTRICAL BSMT-PWR PLAN-OVERALL					\$ -
			Install new power distribution, feeders, and connections to PCW pumps, VFDs, and skid equipment.					\$ -
Electrical	EE.0	5	LCLS II - HE-EI ELECTRICAL SCHEDULES					\$ -
			Provide panel schedules and circuit assignments; supports electrical scope but not standalone construction work.					\$ -
Electrical	E7.2	5	LCLS II - HE-EI SINGLE LINE - MCC 4MCC03-950					\$ -
			Replace breakers, install disconnects, remove existing VFDs, and install/connect new VFDs and wiring to PCW pumps with MCC modifications.					\$ -
Mechanical	M3.0	4	LCLS II - HE-EI BSMT (NEH) - OVERALL PLAN					\$ -
			Install new PCW skid with anchors and isolation, connect new 8" PCWS/PCWR and CHWS/CHWR piping to existing mains, route to sub-basement via existing penetrations, install valves, air separator, filter drain, PLC panel supports, and reconnect makeup water.					\$ -
Mechanical	ME.1	4	LCLS II - HE-EI MECHANICAL DETAILS					\$ -
			Provide and install PCW skid mounting, manifold assemblies, piping, valves, instrumentation, and unistrut supports including anchors, clamps, and wall/ceiling support systems.					\$ -
Mechanical	ME.3	4	LCLS II - HE-EI MECHANICAL DETAILS SCHEMATIC - PCW P&ID					\$ -
			Provide and connect PCW skid instrumentation, valves, air separator, expansion tank, makeup water, and control interfaces per P&ID; coordinate I&EC scope where noted.					\$ -
Mechanical	ME.4	4	LCLS II - HE-EI MECHANICAL MECH SCHEMATIC - PCW FLOW DIA					\$ -
			Install PCW skid system components and piping connections required to meet flow requirements including pumps, filters, and distribution paths per flow diagram.					\$ -
Structural	S3.1	4	LCLS II - HE-EI BSMT - PCW SKID					\$ -
			Provide structural support, anchorage, and mounting system for PCW skid including base plates, anchors, and support framing.					\$ -
3.2 Sub-basement Hallway PCW Upline								
Mechanical	MD2.0	5	LCLS II - HE-EI SUB-BSMT - OVERALL PLAN - DEMO					\$ -
			Demo existing PCWS/PCWR branches and mains, cap lines, provide isolation valves, and coordinate shutdowns and penetrations for future routing. Remove HVAC Ductwork and Light Fixtures for construction accessibility.					\$ -
Mechanical	M2.0	5	LCLS II - HE-EI SUB-BSMT - OVERALL PLAN					\$ -
			Install new 6" PCWS/PCWR mains, 3" branch drops, isolation valves, capped future connections, and route piping through sub-basement corridors with coordination.					\$ -
Mechanical	M2.1	5	LCLS II - HE-EI SUB-BSMT - ENL PLAN - HUTCH 1.3					\$ -
			Provide PCWS/PCWR branch piping, drops, valves, supports, and routing to equipment areas; final connections and manifolds by SLAC.					\$ -
4. Commissioning / Testing / QA								
			System flushing					\$ -
			Pressure testing					\$ -
			Hydronic balancing (TAB)					\$ -
			Startup & commissioning support					\$ -
			QA/QC plan development					\$ -
			Inspection support					\$ -
			Test procedures & reports					\$ -
			As-built & O&M manuals					\$ -
5. Indirect Fees								
			General Conditions					\$ -
			Bonds and Insurance					\$ -
			Overhead and Profit					\$ -
TOTAL BASE SCOPE								\$ -